

(4, X)

Time: 1hr.

Class Test-1 on Computer Graphics & Geometric Modeling, 2026

Set-1

Full Marks: 30

CQ1	a. Given the center point coordinate (20, 16) and a radius of 25 unit, generate all the points (in the 2nd octant) by using both the Bresenham and Midpoint circle drawing algorithms. Also, compare the efficiency of both algorithms in terms of the number of calculations and the computational complexity involved in plotting the points. b. Suppose you have a system with a 12 inch by 14 inch video monitor that can display 120 pixels per inch. If memory is organized in one byte words, the starting frame buffer address is 0, and each pixel is assigned 4 bits of storage, what is the frame buffer address of the pixel with screen coordinates (x, y)?	8+3
CQ2	a. Determine the Bezier blending functions for five control points. Plot each function and label the max and min values. b. Suppose the knot vector of a B-spline is {0, 0, 0, 1, 2, 3, 3, 3}. Find the possible values of number of control points and degree of the polynomial.	7+3
CQ3	a. The lower left-hand and upper right-hand corners of a rectangular clipping window are at (-4, -3) and (2, 2), respectively. Using the Sutherland-Cohen algorithm, check and find the visibility portion of the quadrilateral ABCD for A(-5, -2), B(-2, -1), C(0, 3), and D(1, 1) in the clipping window.	9

Time: 1hr.

Class Test-1 on Computer Graphics & Geometric Modeling, 2026

Set-2

Full Marks

CO1	a. Given the center point coordinate (15, 10) and a radius of 20 unit, generate all the points (in the 3rd octant) by using both the Bresenham and Midpoint circle drawing algorithms. Also, compare the efficiency of both algorithms in terms of the number of calculations and the computational complexity involved in plotting the points. b. How long would it take to load a 640 x 480 frame buffer with 12 bits per pixel, if 105 bits can be transferred per second? How long would it take to load a 24-bit per pixel frame buffer with a resolution of 1280 x 1024 using this same transfer rate?	8+
CO2	a. Given a cubic Bézier curve defined by the control points $P_1(-8, 0, 0)$, $P_2(-4, 5, 0)$, $P_3(4, -5, 0)$, and $P_4(8, 0, 0)$. Determine the geometric coefficient matrix that reproduces this curve in the cubic Hermite form. b. Obtain no. of segments, no. of control points in each segment, total control points, degree and order of the B-spline curve defined by $n=5$ and $k=3$.	7
CO3	a. The lower left-hand and upper right-hand corners of a rectangular clipping window are at (-3, 1) and (2, 6), respectively. Using Sutherland-Cohen algorithm, check and find the visibility portion of the quadrilateral EFGH for $E(-4, 2)$, $F(-1, 7)$, $G(-1, 5)$, and $H(-2, 3)$ in the clipping window.	

Time: 1hr.

Class Test-1 on Computer Graphics & Geometric Modeling, 2026

Set-3

Full Marks: 30

CO1	a. Given the center point coordinate (10, -5) and a radius of 8 unit, generate all the points (in the 4th octant) by using both the Bresenham and Midpoint circle drawing algorithms. Also, compare the efficiency of both algorithms in terms of the number of calculations and the computational complexity involved in plotting the points. b. Consider a non-interlaced raster monitor with a resolution of n by m (m scan lines and n pixels per scan line), a refresh rate of r frames per second, a horizontal retrace time of $t_{\text{horizontal}}$ and a vertical retrace time of t_{vertical}. What is the fraction of the total refresh time per frame spent in retrace of the electron beam.	8+3
CO2	a. A cubic splined curve has start and end point at (20, 0) and (7, 2) respectively. The tangent vectors at start and end points meet at (18, 9). Determine the Hermit cubic spline curve and hence find out its mid-point. b. Define the knot values for $n=3$ and $k=3$ where symbols carry their usual meaning.	7+3
CO3	a. The lower left-hand and upper right-hand corners of a rectangular clipping window are at (50, 10) and (80, 40), respectively. Using Sutherland-Cohen algorithm, check and find the visibility portion of the quadrilateral PQRS for P(35, 10), Q(65, 40), R(65, 20), and S(95, 10) in the clipping window.	9